

Natural Units, the Cosmological Hierarchy, and a Ramanujan–Heegner Numerical Curiosity

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Buckingham’s Π -theorem provides a linear-algebraic account of natural units. The constants c , $\hbar$, and G form a dimensionally independent basis for the mechanical dimensions M , L , and T . Introducing the Hubble parameter H produces one dimensionless invariant, $\hbar GH^2/c^5$. Depending on the chosen natural-unit basis, the observed hierarchy of order 10^{-122} may appear in H^2 , G , or Λ_{P}^2 . A numerical curiosity then follows: the inverse seventh power of the Ramanujan constant $R = e^{\pi\sqrt{163}}$ is approximately 1×10^{-122} . The numbers 7 and 163 are both Heegner numbers, though no physical mechanism is implied.

I. DIMENSIONAL SPACE AND BUCKINGHAM’S THEOREM

Buckingham’s Π -theorem states that a physical relation involving n dimensional quantities constructed from r independent base dimensions may be expressed using $n - r$ independent dimensionless combinations.

For relativistic gravitation, take the mechanical base dimensions to be $[M]$, $[L]$, $[T]$. Represent each physical quantity by its exponent vector in the ordered basis (M, L, T) . The dimensions of the speed of light, the reduced Planck constant, and Newton’s gravitational constant are

$$[c] = LT^{-1}, \quad (1)$$

$$[\hbar] = ML^2T^{-1}, \quad (2)$$

$$[G] = M^{-1}L^3T^{-2}. \quad (3)$$

Their dimensional exponent vectors are therefore

$$d_c = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \quad d_{\hbar} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad d_G = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix}. \quad (4)$$

Using these vectors as columns gives the dimensional matrix

$$D = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 2 & 3 \\ -1 & -1 & -2 \end{pmatrix}.$$

Its determinant is $\det D = -2 \neq 0$. Hence $\text{rank}(D) = 3$, and the three constants c , $\hbar$, and G are dimensionally independent. They form a basis for the three-dimensional mechanical dimensional space. This establishes the linear-algebraic legitimacy of the natural-unit choice $c = \hbar = G = 1$. The constants have not disappeared physically. Their dimensions have been absorbed into the definitions of the units themselves.

II. ADDING THE HUBBLE PARAMETER

The Hubble parameter has dimensions $[H] = T^{-1}$, and hence dimensional vector $d_H = (0 \ 0 \ -1)^T$. Adjoining this

fourth quantity gives

$$D_{\text{ext}} = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 1 & 2 & 3 & 0 \\ -1 & -1 & -2 & -1 \end{pmatrix}.$$

The rank remains three: $\text{rank}(D_{\text{ext}}) = 3$. Since there are four quantities and three independent base dimensions, $\text{nullity}(D_{\text{ext}}) = 1$. Buckingham’s theorem therefore guarantees one independent dimensionless combination. Let $\Pi = c^a \hbar^b G^d H^e$. Dimensional homogeneity requires

$$D_{\text{ext}} \begin{pmatrix} a \\ b \\ d \\ e \end{pmatrix} = \mathbf{0}.$$

A basis vector for the null space is

$$\begin{pmatrix} a \\ b \\ d \\ e \end{pmatrix} = \begin{pmatrix} -5 \\ 1 \\ 1 \\ 2 \end{pmatrix}.$$

The resulting invariant is

$$\Pi = \frac{\hbar GH^2}{c^5}.$$

Using the Planck time,

$$t_{\text{P}} = \sqrt{\frac{\hbar G}{c^5}},$$

this may be written $\Pi = (H t_{\text{P}})^2$. For the present Hubble parameter H_0 , $H_0 t_{\text{P}} \sim 10^{-61}$, and therefore

$$\Pi_0 = \frac{\hbar G H_0^2}{c^5} \sim 10^{-122}.$$

This small dimensionless number cannot be eliminated through a change of units. A change of basis can only determine which dimensional constant carries it.

III. PLANCK AND HUBBLE NATURAL UNITS

In Planck units one chooses $c = \hbar = G = 1$. The invariant then reduces to $\Pi_0 = H_0^2$, so that

$$H_0^2 \sim 10^{-122}.$$

One may instead adopt a Hubble-adapted basis, $c = \hbar = H_0 = 1$. The same invariant then becomes $\Pi_0 = G$, and therefore

$$G_{\{c=\hbar=H_0=1\}} \sim 10^{-122}.$$

This is the same hierarchy expressed in another dimensional basis:

$$H_0^2 \sim 10^{-122} \iff G \sim 10^{-122}.$$

The hierarchy has moved from the expansion scale to the gravitational coupling.

IV. CONNECTION WITH THE COSMOLOGICAL CONSTANT

For a de Sitter spacetime,

$$H^2 = \frac{\Lambda c^2}{3}.$$

The Planck length is

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}.$$

Multiplying the de Sitter relation by $\hbar G/c^5$ gives

$$\frac{\hbar G H^2}{c^5} = \frac{\Lambda \hbar G}{3c^3}.$$

Since

$$\ell_P^2 = \frac{\hbar G}{c^3},$$

it follows that

$$\Lambda \ell_P^2 = 3 \frac{\hbar G H^2}{c^5}.$$

Thus the same hierarchy may be expressed as

$$(H_0 t_P)^2 \sim 10^{-122},$$

$$\Lambda \ell_P^2 \sim 10^{-122},$$

or, in the basis $c = \hbar = H_0 = 1$, $G \sim 10^{-122}$, up to conventional factors of order unity. The cosmological hierarchy is therefore

the invariant content. Its location among H , G , and Λ depends on the chosen dimensional basis.

V. A RAMANUJAN-HEEGNER COINCIDENCE

Define the Ramanujan constant $R = e^{\pi\sqrt{163}}$. Numerically, $R \approx 2.6253741264 \times 10^{17}$. Its seventh inverse power is

$$R^{-7} = e^{-7\pi\sqrt{163}} \quad (5)$$

$$\approx 1.1632 \times 10^{-122}. \quad (6)$$

Hence

$$R^{-7} \approx 10^{-122}.$$

In Hubble-adapted natural units, this may be placed beside

$$G_{\{c=\hbar=H_0=1\}} \sim 10^{-122},$$

giving the numerical comparison

$$G_{\{c=\hbar=H_0=1\}} \sim 10^{-122} \approx R^{-7}.$$

The pair (7, 163) is arithmetically conspicuous because both entries are Heegner numbers. The complete set of Heegner numbers is

$$1, 2, 3, 7, 11, 19, 43, 67, 163.$$

The appearance of 163 in $R = e^{\pi\sqrt{163}}$ is tied to the class-number-one property of the imaginary quadratic field

$$\mathbb{Q}(\sqrt{-163})$$

and to the theory of complex multiplication of elliptic curves. The exponent 7, though itself a Heegner number, has no established cosmological derivation in this setting.

VI. CONCLUSION

Buckingham's theorem identifies the dimensionless combination

$$\frac{\hbar G H^2}{c^5}$$

because four dimensional quantities are being described within a three-dimensional mechanical dimensional space. Natural units correspond to a choice of basis in that space. Choosing $c = \hbar = G = 1$ places the hierarchy in H^2 ; choosing $c = \hbar = H_0 = 1$ places it in G ; writing the same quantity geometrically places it in $\Lambda \ell_P^2$.

The relation

$$R^{-7} \approx 10^{-122}$$

is a numerical nicety attached to this genuine dimensional hierarchy. The Buckingham invariant supplies the physics. The Heegner pair (7, 163) supplies the numerology.